DELIVERY STATION OPERATIONS ANALYTICS

Operational performance, staffing efficiency and dispatch reliability

Analysis window: 01 Jun 2026 – 03 Oct 2026
Source: Delivery_station_main_table.xlsx · 12,000 operational records

01 Executive summary

A management-oriented view of throughput, labor, dispatch and operational quality.

5.77M Packages processed	**7.40M** Packages received	**48.4** Packages per labor hour	**28.6%** Rows flagged on-time dispatch
193 Average recorded backlog	**6.9%** Rows below planned staffing	**62,345** Recorded missorts	**6,859** Overtime hours recorded

- Throughput totals 5,765,681 packages against 7,404,116 received; the difference indicates that receipt and processing measures need to be interpreted alongside backlog and flow timing.
- Recorded dispatch actuals are 5,983,670 versus a target of 7,070,212; validate whether these fields are comparable and aligned to the same operational unit/time window.
- The dataset is suitable for exploratory operations reporting, but KPI definitions should be confirmed before decisions or external reporting.

02 Data scope & quality

The workbook contains a single operational fact table. The checks below describe the file as supplied.

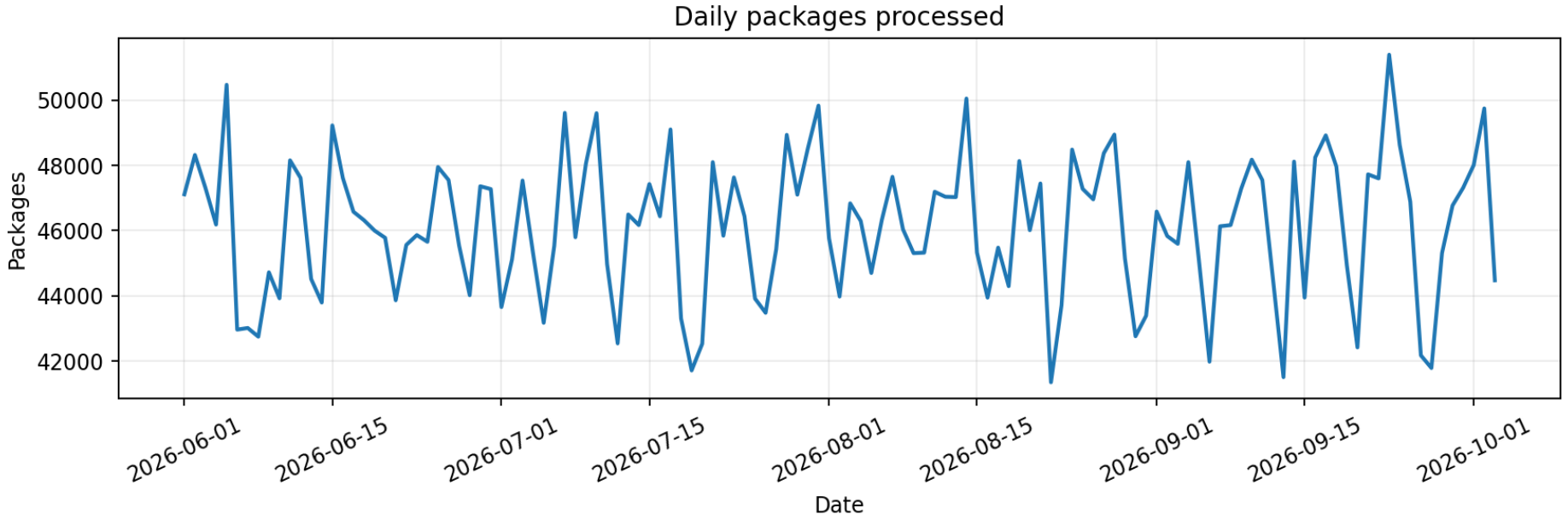
Data coverage	12,000 rows · 12,000 unique operation IDs
Date range	01 Jun 2026 to 03 Oct 2026 (125 distinct dates)
Missing values	0 across all columns
Duplicate rows	0 exact duplicate rows
Fields	Date/hour, shift, process, work area, received/processed packages, backlog, staffing, labor, dispatch, missorts, problem packages and overtime.

Important validation flags

- On_Time_Dispatch averages 28.6% when treated as a 0/1 indicator. Confirm that 1 means on-time and that the field is populated consistently.
- Dispatch_Actual totals 5,983,670, materially below Dispatch_Target at 7,070,212. Confirm aggregation grain, units and whether target/actual refer to the same time window.
- Backlog is treated here as a recorded row-level value; do not sum it across rows unless the data model confirms that each row represents a non-overlapping backlog amount.

03 Throughput over time

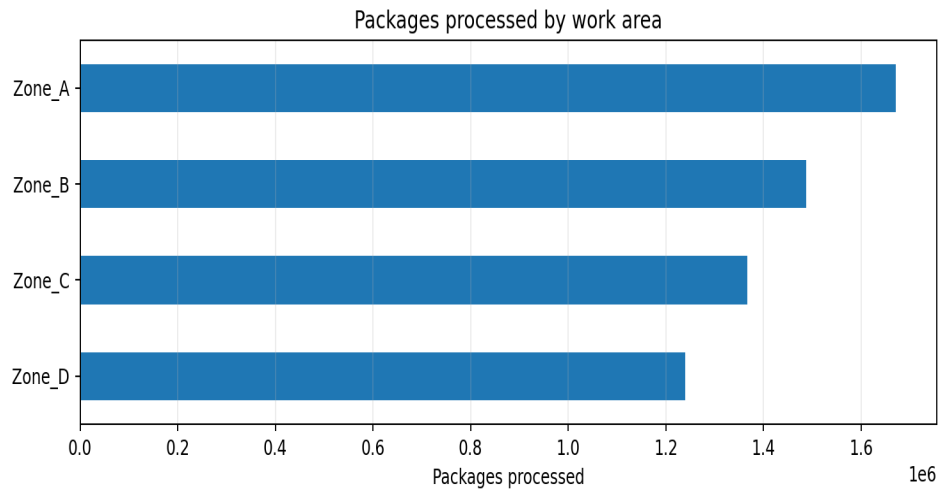
Daily processed volume shows the operating rhythm and helps highlight peak days or throughput dips.

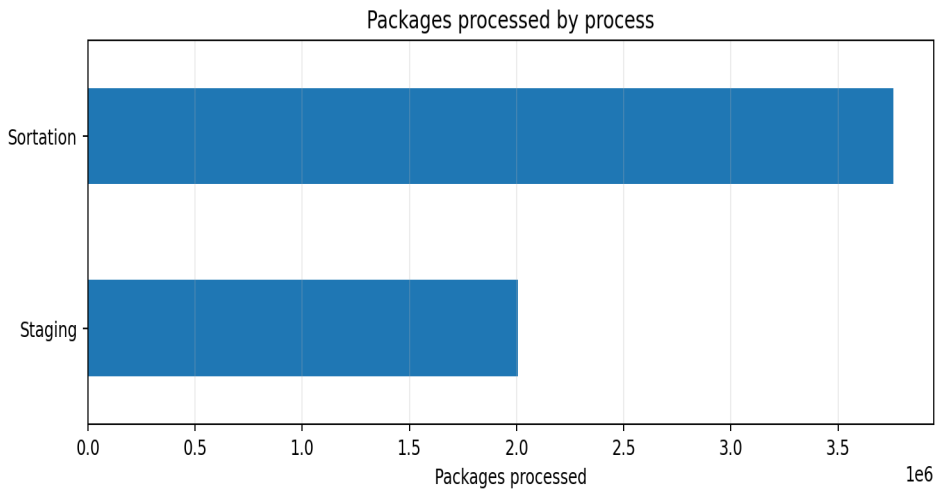


- Average daily processed volume: 46,125 packages; lowest day: 41,357; highest day: 51,384.
- Use daily trend exceptions to review inbound volume, staffing, equipment availability, and dispatch cut-off constraints together.

04 Work-area & process mix

Volume is not evenly distributed across work areas and process types.

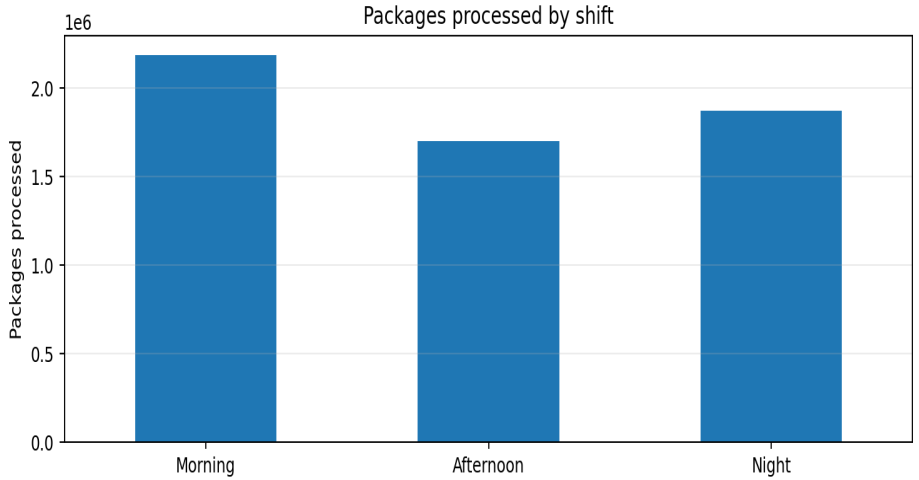


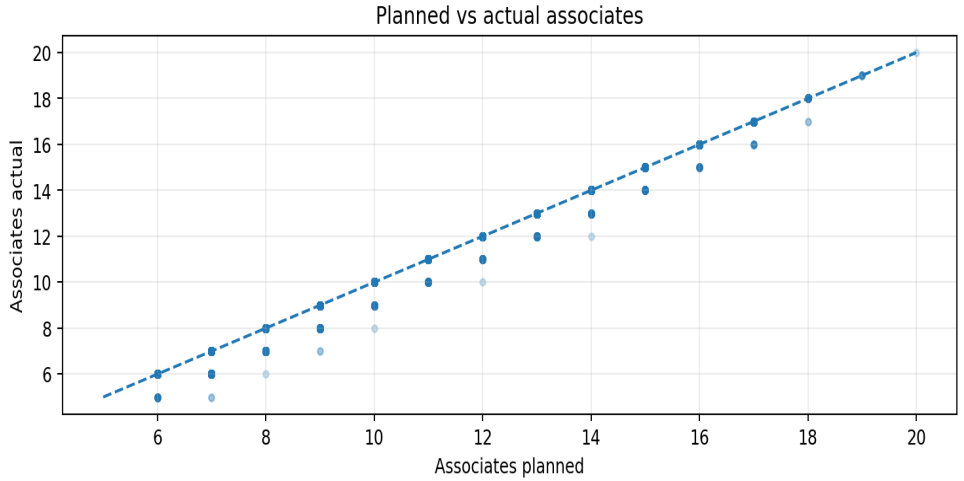


- Zone_A contributes the largest recorded work-area volume (1,671,538 packages; 29.0% of processed volume).
- Sortation accounts for 3,758,453 packages (65.2%); compare staffing and productivity targets separately by process.

05 Shift profile & staffing

Compare shift workload and identify where planned staffing is not met.

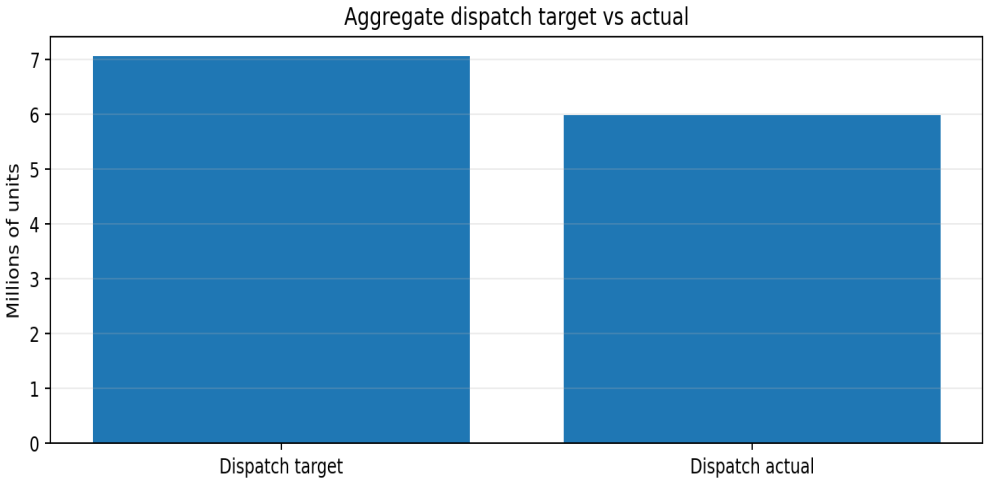




- Across all rows, planned associate counts total 120,098 versus 119,258 actual: a net gap of 840 associate-records (not necessarily unique people).
- 6.9% of records have actual staffing below planned staffing. Check the gap by shift, hour, process and work area before changing roster buffers.
- The scatter plot’s dashed line marks equal planned and actual staffing; points below it represent under-staffed records.

06 Dispatch & operational quality

Dispatch performance and quality indicators deserve a definition check before being treated as final KPIs.



Recorded missorts **62,345**

Problem packages **36,690**

Dispatch actual vs target **84.6% of target**

Overtime recorded **6,859.3 hours**

- Aggregate dispatch actual is 1,086,542 below target. Investigate whether this reflects true missed dispatch, different metric definitions, or mismatched aggregation grain.
- Missorts and problem packages should be normalized (for example, per 1,000 packages processed) and segmented by hour, process and work area to make comparisons fair.

07 Recommendations & action plan

Turn exploratory charts into a repeatable management operating rhythm.

1. Validate KPI definitions	Confirm row grain, backlog meaning, dispatch target/actual units, and On_Time_Dispatch coding with station leadership.
2. Build a daily scorecard	Track packages received/processed, backlog trend, units per labor hour, on-time dispatch, missort rate, problem-package rate and overtime.
3. Optimize staffing by workload	Compare planned vs actual staffing by shift × hour × process × work area; test a small attendance buffer only where historical gaps justify it.
4. Investigate exceptions	Flag days with low throughput, high backlog, high missorts, or dispatch variance and pair each exception with a root-cause note.
5. Operationalize the dashboard	Use Python for data preparation and validation, SQL for reproducible KPI logic, and Power BI for a manager-facing view with date/shift/area filters.

Suggested next deliverable: a Power BI dashboard with an executive overview, throughput & backlog, staffing & productivity, and dispatch & quality pages.

Source note: All figures and charts in this presentation are calculated from the uploaded workbook. Findings are descriptive, not causal; validate metric definitions before operational decisions.